

Bio-Inspired Morphing-Wing Control: Curriculum Soft Actor-Critic Learning and KAN-Based Policy Distillation

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Abstract

Micro air vehicles (MAVs) enable the autonomous operation of essential missions such as search and rescue, disaster monitoring, and smart-city infrastructure inspection, where the cramped spaces and cluttered environment deter the use of traditional large-winged aircraft. However, the use of discrete surfaces such as ailerons to command directional moments such as yaw results in redundant mechanisms that incur unnecessary mechanical weight and complexity. This paper addresses yaw control of a bio-inspired morphing-wing glider that uses no vertical stabilizers, relying on a compliant cubic Bézier spar as the sole actuator. Then, a six-component simulation stack is presented, which includes cubic Bézier energy solver, quasi-steady strip-theory aerodynamic proxy, full 6 degree of freedom (DOF) quaternion dynamics and Curriculum Soft Actor-Critic (SAC). The SAC achieves a $4.4\times$ accuracy improvement over the best classical controller, with a 2% failure rate. Curriculum SAC is statistically equivalent, and the interpretability of curriculum SAC reveals an unexpected attitude-first control priority. In summary, the Policy Interpretability Suite paves the way for a more efficient and safer approach to bio-inspired aircraft control, in particular, the morphing-wing glider.

Keywords: Morphing wing, Soft actor-critic, Curriculum learning, Yaw rate tracking, Policy distillation

1 Introduction

Micro air vehicles (MAVs) have been extensively studied due to their ability to navigate cramped and cluttered environments where conventional large-winged aircraft fail, enabling the autonomous operation of critical missions like collaborative search and rescue, disaster monitoring, and smart-city infrastructure inspection [1–5]. However, conventional MAVs rely on discrete control surfaces, such as ailerons, to command directional forces, resulting in redundant mechanisms prone to jamming, increased mass, and accumulated mechanical debt [6–8]. Yaw authority — the ability for an aircraft to steer its direction — is less commonly discussed.

Nearly all reinforcement learning (RL) flight control studies assume a rigid body, with the presence of discrete control surfaces, such as ailerons, being extremely prevalent [9]. A key advantage of such a setup is the linearization of actuation and resulting aerodynamic forces, with drawbacks being added mass and hardware complexity. In fact, there have been no studies that employ deep reinforcement learning agents to control a fully compliant morphing spar with a central focus on yaw authority — a possible explanation being that real-time Finite Element Analysis (FEA) is too computationally prohibitive for millions of RL timesteps [10].

The viability of Deep Reinforcement Learning has consistently been demonstrated in prior UAV works [9, 11]. For instance, Böhn et al. showed that Proximal Policy Optimization (PPO) was able to modulate attitude for fixed-wing aircraft [11]. However, to address the highly nonlinear morphing wing problem, the concept of Soft-Actor-Critic (SAC) reinforcement learning algorithm and its max-entropy objective is critical [12]. Maximizing entropy encourages agent exploration and prevents the premature convergence of its policy, characteristics vital for discovering the sweep-dihedral coupling from scratch [12]. Additionally, SAC’s sample efficiency, stemming from its off-policy replay, reduces training time and ensures the feasibility of running millions of environmental steps on a single device [12].

To bridge the previously mentioned gap in research and employ the morphing wing effectively, the glider’s yaw is controlled through the following control sequence: the RL agent provides changes in the position of the two P_3 coordinates, the Bézier curve energy minimization algorithm solves for the newly arising shape, the implicit twist is activated, creating geometric washout, and strip theory integrates the aerodynamic forces to compute the final yaw moment. Since no linearization is possible around the neutral configuration, an SAC algorithm is introduced, treating the RL agent’s control policy initially as a black-box. The high FEA computational cost is addressed by imposing a warm-start condition on the wing-shape solver, forcing the energy minimization algorithm to solve for the new wing curvature from the previous timestep’s solution. As a result, the duration per timestep for the proposed Bézier proxy can be dramatically reduced to 10 iterations per 40ms step, allowing the RL algorithms to solve for the updated Bézier curvatures within microseconds.

Five classical controllers are subsequently benchmarked for comparison: a virtual tendon heuristic, Proportional-Integral-Derivative (PID), Gain-Scheduled PID, Linear Quadratic Regulator (LQR), and Linear Model Predictive Control (MPC). Each of these methods, in some form, linearizes the plant around a trim condition where the sweep-dihedral product is zero — the exact configuration where yaw authority

vanishes. As a result, they cannot exploit the multiplicative nonlinearity that the agent must discover as these controllers are fundamentally built on the assumption that individual control inputs translate to separable aerodynamic effects. The experiments in Section IV confirm this empirically: all five achieve 96–100% episode failure at the 65° roll/pitch evaluation limit, which is a structural consequence of the non-separable kinematics rather than a failure of controller tuning.

After policy learning, the RL agent’s yaw control strategy is subsequently elucidated through a series of interpretability suites. This process includes invoking finite-difference Jacobian feature importance, latent-space Principal Component Analysis (PCA), and Dataset Aggregation (DAgger) [13] to distill the dense policy into a compact Kolmogorov-Arnold Network (KAN) [14]. Though slightly sophisticated, this process guarantees that the agent has discovered a generalized method that is grounded in physics instead of memorizing a set of arbitrary actions.

The main contributions of this work are as follows:

1. Bézier Spar Energy Solver: A cubic Bézier curve and an energy minimization solver are presented to model each wing of the glider, a lightweight approach that efficiently models gull-like curvature at high control frequencies.
2. Quasi-Steady Strip-Theory Aerodynamic Proxy: A Quasi-Steady Strip-Theory panel model is presented, with 6 panels during training and 10 during evaluation. By summing the yaw moment per panel across the span, calculated through effective local angle-of-attack, body rates and gust, the presented model is an effective aerodynamics proxy that bypasses complex Computational Fluid Dynamics (CFD) models. The robustness of the learning policy, which utilizes the aerodynamics model, is subsequently preserved through the domain randomization of lift_scale $\pm 30\%$, spanning the entire empirical range.
3. Six-Phase Curriculum Framework: A 6-phase ProgressiveTwistWrapper curriculum is proposed, gradually relinquishing dihedral authority and domain randomization to the curriculum agent. This method presents the agent with a simpler, more predictable, lower-authority problem while mitigating the potential for direct SAC to converge to $z = 0$. However, notably, flat SAC, with the same total timestep budget, matches curriculum performance (Wilcoxon $p = 1.0$), a null result reported and discussed in Section V.
4. Residual Heuristic Learning (Negative Result): A Residual-Heuristic approach is developed, where the main driver of yaw control is a heuristic manually designed proportional-damping controller with one machine-teaching-extracted feedforward term. The SAC’s objective is to add a corrective residual term on top of the heuristic controller, with the intent to accelerate learning as the agent is starting from a decent baseline instead of random noise.
5. Policy Interpretability Suite: The suite comprises finite-difference Jacobian feature importance (top feature: `cos_pitch` = 0.608, revealing attitude-first priority over `yaw_ref` = 0.501), latent-space Principal Component Analysis (PCA)/ t-distributed Stochastic Neighbor Embedding (t-SNE) — PC1 separates yaw target sign; policy spends 40.5% of timesteps in a stressed regime; DAgger distillation to

a KAN student (33,240 parameters, $2.3\times$ compression, 0.259 RMS@H), and cross-agent action channel comparison confirming both baseline and curriculum SAC independently converge to differential dihedral modulation — a task-determined strategy that mechanistically explains the Wilcoxon $p = 1.0$ null result.

2 System Architecture

The simulation stack comprises five coupled components: a cubic Bézier spar geometric model, a quasi-steady strip-theory aerodynamic proxy, a full 6 degrees of freedom (DOF) quaternion dynamics, domain randomization, and a Gymnasium MDP environment. Each module is individually validated, allowing ablation of individual modules while keeping the overall pipeline end-to-end differentiable for RL training. The framework is described from the bottom up, in the sequence of wing geometry, flight dynamics and subsequently the MDP formulation used for training.

2.1 Wing Geometric Model

The spar shape is defined as:

$$B(u) = (1 - u)^3 P_0 + 3(1 - u)^2 u P_1 + 3(1 - u) u^2 P_2 + u^3 P_3 \quad (1)$$

Each wing spar is modelled as a cubic Bézier curve with four control points P_0 through P_3 , parameterized by $u \in [0, 1]$ along the arc length [15]. P_0 is root-fixed to the fuselage; P_3 is the agent-commanded wingtip with bounds $\Delta x \in [-0.45, 0.45]$ m and $\Delta z \in [-0.15, 0.15]$ m. The interior points P_1 and P_2 are not directly controlled; they are repositioned at each step by the energy minimization solver described below. The choice of cubic over quadratic provides C^2 continuity, two free interior degrees of freedom, and the ability to represent the inflection point required for a gull-wing curvature, all while keeping the agent’s action dimensionality to 3D per wing.

Compared with a quadratic-degree curve, the third-degree curve can increase the plausible gull-wing postures while restricting complexity, ensuring that each timestep is computationally feasible. The solver repositions the internal points P_1 and P_2 at 25 Hz, with an emphasis on naturalistic smooth curvature. The fixed-length constraint and bio-mimicry bias are enforced, rewarding ‘rising’ internal points P_1 and P_2 as seen in (2) to achieve more ‘n’-shaped, gull-wing postures without restricting the agent’s freedom in manipulating P_3 ’s position. The warm-start policy is also imposed, requiring the energy minimization algorithm to reuse the solution (points P_1 and P_2) in the previous timestep, reducing the number of iterations required for the convergence of the warm-start’s gradient descent to only about 10 iterations per control step. It tracks Euler–Bernoulli bending energy extremely closely ($r = 0.9998$) and executes in microseconds—speeds fast enough to support millions of RL steps.

$$J = w_{\text{energy}} \sum_i |P_i - P_{i-1}|^2 + w_{\text{len}} (L - L_{\text{fixed}})^2 - w_{\text{bio}} (z_{P_1} + z_{P_2}) \quad (2)$$

The objective is minimized by finite-difference gradient descent with a learning rate of 0.04 and gradients clipped to 0.6 per axis, running for approximately 10 iterations

per 40 ms control cycle. Rather than reinitializing from scratch at each step, the solver begins from the previous timestep’s P_1 and P_2 positions, enforcing temporal continuity and dramatically reducing the iterations needed for convergence.

$$\vec{c}(u) = (1 - u)\hat{i}_{\text{body}} + u \left(\hat{\mathbf{T}}(u) \times \hat{\mathbf{k}} \right) \quad (3)$$

The chord vector is an interpolation between the root chord, aligned with the glider body’s x -direction, and the tip chord, perpendicular to the local Bezier tangent. When the tip chord rotates relative to the freestream, due to the non-zero angle of attack and a non-zero sweep angle, the cross product of the y -axis unit vector and the tip chord vector results in a vector that both points in the same direction as sweep and lies perpendicular to the y -direction. As shown in Fig. 1, which illustrates the ΔM_z (yaw moment authority) heatmap over d_x vs d_z , the multiplicative effect of having both dihedral and sweep still holds here — the 0 ΔM_z at $d_x = 0$ or $d_z = 0$ makes the relationship of zero-crossing extremely explicit.

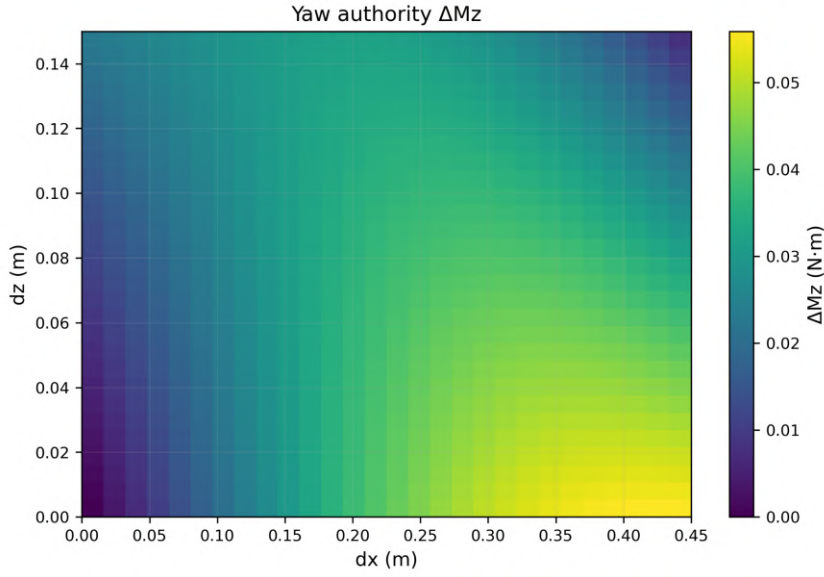


Fig. 1 Yaw moment authority ΔM_z as a function of right-wing tip sweep offset dx and dihedral offset dz . The zero-crossing at d_x or d_z illustrates the multiplicative nonlinearity: yaw authority requires simultaneous non-zero sweep and dihedral.

2.2 Aerodynamic Model

Each wing spar is discretized into N panels ($N = 6$ during training, $N = 10$ during evaluation) along the spanwise direction. For each panel i , the local effective angle of attack α_i is computed from the full relative velocity vector, accounting for body rotation rates and atmospheric gust, projected onto the local panel normal. This ensures

that gyroscopic coupling between roll rate and effective lift angle is captured, which is important for the attitude-stabilization dynamics. Panel forces are integrated numerically and moments summed about the body center of mass to produce the aerodynamic torque vector M_{aero} .

Each panel i at spanwise coordinate s_i contributes lift and drag according to the thin-airfoil approximation with a post-stall flat-plate fallback:

$$C_{L,i} = \text{clip!}(c_{L\alpha} \alpha_i, -C_{L,\max}, C_{L,\max}) \quad (4)$$

$$C_{D,i} = c_{d0} + k_{\text{ind}} C_{L,i}^2 \quad (5)$$

where α_i is the local effective angle of attack (including body rates and gust velocity), $c_{L\alpha} = 4.5 \text{ rad}^{-1}$, $C_{L,\max} = 1.4$, $c_{d0} = 0.02$, and $k_{\text{ind}} = 0.04$. This proxy is calibrated with a nominal effective lift slope 4.25 rad^{-1} — approximately 31% below the horse-shoe inviscid Vortex Lattice Method (VLM) benchmark (the VLM reference assumes local angle-of-attack (AoA) α fulfills the condition $\alpha \in [-15, 15]$ where most flow stays attached. The glider stays well within this range during normal flight). Validation against that VLM benchmark yields a Pearson’s correlation coefficient $r = 0.998$ and RMSE = 0.270 N, suggesting that the proxy has the correct functional form, with the remaining slope deviation stemming from the viscous boundary-layer effects at $\text{Re} \approx 10^5$.

$$\mathbf{I} \dot{\boldsymbol{\omega}} = \mathbf{M}_{\text{ar}} - \text{diag}(d_r, d_p, d_y) \boldsymbol{\omega} - \boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega}) \quad (6)$$

Rotational dynamics are modelled in (6), where the inertia tensor is set to be $\mathbf{I} = \text{diag}(0.08, 0.10, 0.15) \text{ kg}\cdot\text{m}^2$. Rotational drag effects are also taken into account, with the rotor drag coefficients $d_r = 0.3$, $d_p = 0.25$, and $d_y = 0.35 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$. The term $\boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega})$ represents the gyroscopic torque induced on the glider required to maintain its current rotation, which produces unphysical precession during rapid attitude changes.

A full 6DOF aerodynamics model is employed here, as asymmetrical wing configurations inherently induce coupled roll effects due to dihedral changes. As a result, the agent has to learn to both balance attitude control and learn effective yaw tracking without an explicit decomposition. Episode termination limits will be enforced if roll or pitch increases to above 65 degrees or if airspeed drops below stall thresholds, limiting episodes to just 200 steps. The high failure rate of the residual curriculum agent (residual SAC) and all classical controllers is attributed to these exact episode termination mechanisms.

Attitude is tracked via unit quaternion $q = [q_w, q_x, q_y, q_z]^\top$, which avoids the gimbal-lock singularities of Euler-angle representations. Its kinematics are governed by

$$\dot{q} = \frac{1}{2} q \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}, \quad (7)$$

where $\otimes$ denotes the Hamilton product. Both (6) and (7) are integrated forward with an Euler step of $t = 0.04 \text{ s}$. To prevent the glider agent from abusing instantaneous actuator dynamics and making extreme sharp wingtip movements between timesteps that would have accentuated the sim-to-real gap, the actuator lag term is introduced.

By filtering the full wingtip action space (6 positions commands, 3 per wing) through a first-order actuator lag (8) before being applied to the spar, the bandwidth limitations of a servo system can be effectively mimicked. The latency filter can be expressed as:

$$x_{\text{act}}[t+1] = x_{\text{act}}[t] + \alpha_{\text{act}}(x_{\text{cmd}}[t] - x_{\text{act}}[t]), \quad (8)$$

$$\alpha_{\text{act}} = 1 - e^{-\Delta t/\tau} \quad (9)$$

where $\tau = 0.07$ s gives $\alpha_{\text{act}} \approx 0.436$ per step. The policy must therefore anticipate the lag and issue commands 2–3 steps ahead of the desired wing shape.

2.3 Domain Randomization

To improve robustness across the sim-to-real gap, sixteen physics parameters are randomized at the start of each training episode and scaled by the phase-dependent `rand_scale` factor. In the curriculum, `rand_scale` ramps from 0.0 in Phase 1 to 1.0 in Phases 5–6, so the agent first learns in a clean, deterministic environment before facing the full range of physics uncertainty. The flat baseline SAC uses `rand_scale` = 1.0 throughout. Under full randomization, baseline SAC degrades by only 0.046 rad/s RMS@H, confirming that the policy generalizes well across the randomized parameter range.

Domain randomization is applied by sampling all physics parameters uniformly within their bounds at the start of each training episode. Such randomizations cover aerodynamics, inertia, damping, sensor noise, and wind gust effects. Classified as the most consequential perturbations, aerodynamic variations include a $\pm 30\%$ variation in $c_{L\alpha}$ and $\pm 20\%$ in lift scale, spanning the full realistic range of viscous effects at $\text{Re} \approx 10^5$ while simultaneously complementing the more conservative nominal calibration. The wind gusts are modeled as an Ornstein-Uhlenbeck process, with a time constant $\tau_g = 2.0$ remaining unchanged across all three axes of the Cartesian coordinate system to create realistic atmospheric turbulence and effects involved in low-altitude outdoor flight. Table 1 lists all randomized parameters and their perturbation ranges at `rand_scale` = 1.0.

Table 1 Domain Randomisation Parameters
(`rand_scale` = 1.0)

Group	Parameter	Range
Aerodynamic	Lift slope $c_{L\alpha}$	$\pm 30\%$
Aerodynamic	Lift scale	$\pm 20\%$
Aerodynamic	zero-lift drag c_{d0}	$\pm 25\%$
Aerodynamic	induced drag k_{ind}	$\pm 25\%$
Inertia	I_{xx}, I_{yy}, I_{zz}	$\pm 15\%$
Sensor Noise	IMU rate / angle noise	Gaussian
Sensor Noise	Airspeed noise	Gaussian
Wind Gust	O-U process $\tau_g = 2.0\text{s}$	All 3 axes

2.4 MDP Formulation

Table 2 details the forty-one-dimensional observation vector given to the agent. The model, split by key components such as attitude and wing geometry, allows the agent to gain full visibility of its wings’ state configurations contextualized within the global axis. As a result, the previously complex morphological reasoning task is transformed into more manageable pieces with ample quantifiable feedback, a framework that’s potentially able to overcome the agent’s lack of knowledge about explicit aerodynamic domain laws.

Table 2 41-Dimensional Observation Space

Group	Contents	Dim
Attitude	$\sin \phi, \cos \phi, \sin \theta, \cos \theta, \sin \psi, \cos \psi$	6
Body rates	p, q, r (rad/s)	3
Translational	v_x, v_y, v_z , altitude, airspeed	5
Reference	$\psi_{\text{ref}}, e_{\text{yaw}}$	2
Wing geometry	P_1, P_2, P_3 actual & cmd, right & left wings	24
Misc	step fraction t/T_{MAX}	1
Total		41

Table 3 Action Space Bounds (Physical Clamp Limits)

Channel	Min	Max
$\Delta x_{R,L}$ (Sweep)	−0.45 m	+0.45 m
$\Delta y_{R,L}$ (Span)	Fixed (span constraint)	Fixed (span constraint)
$\Delta z_{R,L}$ ¹ (Dihedral)	−0.15 m	+0.15 m

¹ Δz scaled by `twist_factor` $\in [0, 1]$ during curriculum.

$$r = r_{\text{surv}} + r_{\text{track}} - r_{\text{att}} - r_{\text{rates}} - r_{\text{ctrl}} - r_{\text{jerk}} - r_{\text{power}} - r_{\text{struct}} - r_{\text{wall}} \quad (10)$$

Additionally, the agent’s action space bounds are detailed in Table 3. Notably, due to the wing span constraint, $\Delta y_{R,L}$ is fixed to maintain a realistic representation of the wings. The ProgressiveTwistWrapper in the curriculum design of Section III also controls the dihedral channel $\Delta z_{R,L}$ to facilitate agent learning and implements the gating of `twist_factor`.

The nine-term reward in (10) balances three competing objectives: yaw tracking, attitude stability, and wing structural integrity. The tracking and survival terms pull the policy toward meaningful flight, while the remaining seven penalty terms prevent it from achieving that through destructive means. A key tension exists between r_{struct}

and r_{track} : the structural penalty penalizes large dihedral displacements — exactly the motion required to generate yaw authority. This conflict is the primary motivation for the curriculum framework of Section III. The survival bonus $r_{\text{surv}} = +0.5$ per step incentivises the policy to keep the episode alive. The tracking reward uses a Gaussian kernel centered on zero error:

$$r_{\text{track}} = w_{\text{track}} \exp(-k e_{\text{yaw}}^2), \quad k = 20, \quad (11)$$

where $e_{\text{yaw}} = \dot{\psi} - \dot{\psi}_{\text{ref}}$ in rad/s. The attitude penalty applies a dynamically scaled quadratic:

$$r_{\text{att}} = c_{\text{att}}(\phi_{\text{lim}})(\phi^2 + \theta^2), \quad c_{\text{att}} = \frac{90^\circ}{\phi_{\text{lim}}}, \quad (12)$$

so that tighter roll/pitch limits ($\phi_{\text{lim}} = 65^\circ$ final evaluation) automatically increase the attitude penalty weight. Structural cost r_{struct} is proportional to the wing bending energy integral in (2), which suppresses large dihedral displacements early in training and motivates the curriculum. The wall term r_{wall} grows exponentially as $|\phi|$ or $|\theta|$ approaches ϕ_{lim} , providing a soft boundary before hard episode termination. The yaw reference $\dot{\psi}_{\text{ref}}$ is drawn uniformly from the set $\{-0.6, -0.3, 0.0, +0.3, +0.6\}$ rad/s at the start of each segment, held for 50–80 steps, then stepped to a new value drawn from the same set.

3 Curriculum Learning Framework

Classical baseline SAC, without a curriculum structure, is hypothesized to converge to a local optimum, where dihedral (Δz) equals zero, with no yaw authority due to the inherent conflict between the structural energy penalty r_{struct} and the roll/pitch rate penalty r_{rates} both penalizing dihedral (Δz) motion during the first few thousand training steps as described previously. Due to the lack of gradient signal at $\Delta z = 0$, stemming from the tracking gradient $\partial r_{\text{track}} / \partial \Delta z$ being identically zero at $\Delta z = 0$, the agent is unable to escape from this local optimum regardless of additional training steps. The curriculum learning framework mitigates this exact issue by sequencing the available action space and reward in phases such that it allows the agent to build learning momentum from the previous phases and overcome that local optimum.

The curriculum features phase-specific RMS and failure rate criteria, ensuring that the agent-discovered policy isn’t fragile: policies that track well but frequently crash, or those that never crash but are essentially zero-action, are both undesirable. Additionally, phase-specific suppression of structural energy penalty, with earlier phases featuring higher suppression, encourages the agent to explore large wing deformations and arrive at more effective asymmetric dihedral wing configurations.

The ProgressiveTwistWrapper is a Gymnasium action wrapper that scales the agent’s Δz_R and Δz_L outputs by the phase-specific `twist_factor` before passing the command to the environment. The agent always outputs a full 6-dimensional action; only the applied dihedral authority changes across phases. At `twist_factor = 0`, dihedral is fully suppressed and the policy learns from sweep alone. Because the same policy network is reused across phase transitions without re-initialization, the sweep strategy acquired in Phase 1 warm-starts subsequent phases rather than being discarded.

$$r_{\text{twist}} \propto \text{clip!}(-e_{\text{yaw}} \cdot z_{\text{asym}}, -0.1, 0.1) \quad (13)$$

To advance to a subsequent phase, two conditions have to be met simultaneously: the mean evaluation RMS value must fall below a target ρ_k^* , and failure rate must also be below a certain phase threshold f_k^* . The satisfaction of both conditions concurrently is necessary to prevent agents that are either unsustainable, meaning that they track well but crash frequently which invalidates the sim-to-real application, or zero-action, with their failure rate being extremely low but yaw-tracking being statistically identical to zero-action agents.

If performance plateaus — defined as less than 1% improvement in the rolling 10-evaluation window for three consecutive windows — the scheduler temporarily increases the SAC entropy coefficient to encourage exploration, and discards the oldest 30% of the replay buffer to reduce the influence of stale experience. Table 4 summarizes the phase-specific learning rates (LR): Before the learning rate monotonically decays to 1×10^{-4} in the raw_finetune phase, it reaches a peak at approximately 5×10^{-4} during phase 2, which is due to the addition of dihedral in the action space that causes a large change in the agent’s gradient landscape.

Table 4 Six-Phase Curriculum Progression Parameters

Phase	Twist	Rand	RMS tgt	LR
basic_yaw	0.00	0.00	0.52	3×10^{-4}
partial_twist	0.30	0.30	0.38	5×10^{-4}
intermediate_twist	0.45	0.45	0.33	4×10^{-4}
moderate_twist	0.60	0.60	0.28	4×10^{-4}
full_twist	1.00	1.00	0.2	2×10^{-4}
raw_finetune	1.00	1.00	—	1×10^{-4}

The third training paradigm, named ResidualHeuristicWrapper, is evaluated, in which the VirtualTendonHeuristicController provides a base wing action and SAC is trained to output a clipped residual correction added on top. The heuristic is a manually designed proportional-damping controller with a machine-teaching-extracted Δz feedforward slope of -0.1408 m/(rad/s). This approach yields a 90.8% episode failure rate under nominal evaluation and 77.7% under domain randomization. Among surviving episodes, however, the steady-state RMS is 0.140 rad/s — the best tracking performance of any controller tested, confirming that the underlying policy logic is sound when attitude remains stable. The failure mechanism is explored in Section V.

4 Experiments and Results

4.1 Experimental Setup

All policies are trained with the Stable-Baselines3 SAC implementation using SubprocVecEnv (10 parallel workers, fork start method) and VecNormalize observation

normalization. Training uses 5 independent seeds (7, 107, 207, 307, 407) with 50 evaluation episodes per seed per condition. Performance metrics are RMS tracking error over the full horizon (RMS@H), RMS in steady state (RMS_{ss}), episode failure rate (roll or pitch exceeding 65°, or stall), and mean settling time. Statistical analysis uses BCa bootstrap 95% CI ($N = 6000$ resamples), hierarchical two-level bootstrap (seed outer / episode inner), Wilcoxon signed-rank tests with Holm-Bonferroni correction ($\alpha = 0.05$), and Cohen’s d . Note on power: with $n = 5$ seeds the achieved power is 0.20 for a medium effect size ($d = 0.5$); the minimum detectable effect at 80% power is $d \approx 1.75$.

4.2 Controller Performance

As shown in Fig. 2, all five classical controllers — heuristic, PID, LQR, MPC, and GS-PID — achieve 96–100% episode failure at the 65° roll/pitch evaluation limit. Their RMS@H values cluster tightly between 0.837 and 0.872 rad/s, with zero-action performing better, confirming that the failure is structural rather than a matter of tuning. All Holm-Bonferroni corrected Wilcoxon tests within the classical group yield $p < 2 \times 10^{-15}$, but in the wrong direction — PID, LQR, MPC, and GS-PID are each significantly worse than the heuristic, not better. The heuristic remains the best classical baseline at 0.837 rad/s, still a $4.4\times$ gap from baseline SAC.

As shown in Fig. 3, the effect size against the heuristic is Cohen’s $d = -9.19$, a large separation well beyond any ambiguity. Under full domain randomization, performance degrades to RMS@H = 0.237 [0.209, 0.270], a +0.046 rad/s increase (+24% relative), with failure rate rising to 6% — a moderate and acceptable generalization gap. Fig. 4, Fig. 5 and Fig. 6 show that baseline SAC achieves RMS@H = 0.191 [95% CI: 0.170, 0.225] under nominal physics, with a 2% failure rate and a mean settling time of 76.7 steps.

Curriculum SAC achieves RMS@H = 0.299 [0.168, 0.546] with a 21.1% failure rate. The wide confidence interval reflects high seed sensitivity: some seeds converge as well as baseline, while others plateau higher, suggesting that phase transitions occasionally destabilise individual training runs. Despite this, curriculum SAC is not statistically distinguishable from baseline SAC (Wilcoxon $p = 1.0$, $d = -0.56$), a result that is consistent with the low statistical power at $n = 5$ seeds. It is therefore concluded that the curriculum does not measurably improve performance within this experiment’s scope.

Residual SAC’s RMS@H of 0.614 [0.568, 0.656] is dominated by the 90.8% of episodes that terminate early from attitude divergence and should not be treated as a tracking metric. Among the 9.2% of episodes that survive to steady state, RMS_{ss} = 0.140 rad/s — the best of all controllers.

4.3 Curriculum Learning Progression

Figure 7 shows RMS@H over training steps for all three algorithms. Baseline SAC (blue) descends monotonically from approximately 0.82 rad/s to 0.19 rad/s by 2M steps, with a tight BCa CI throughout — a clean, low-variance convergence.

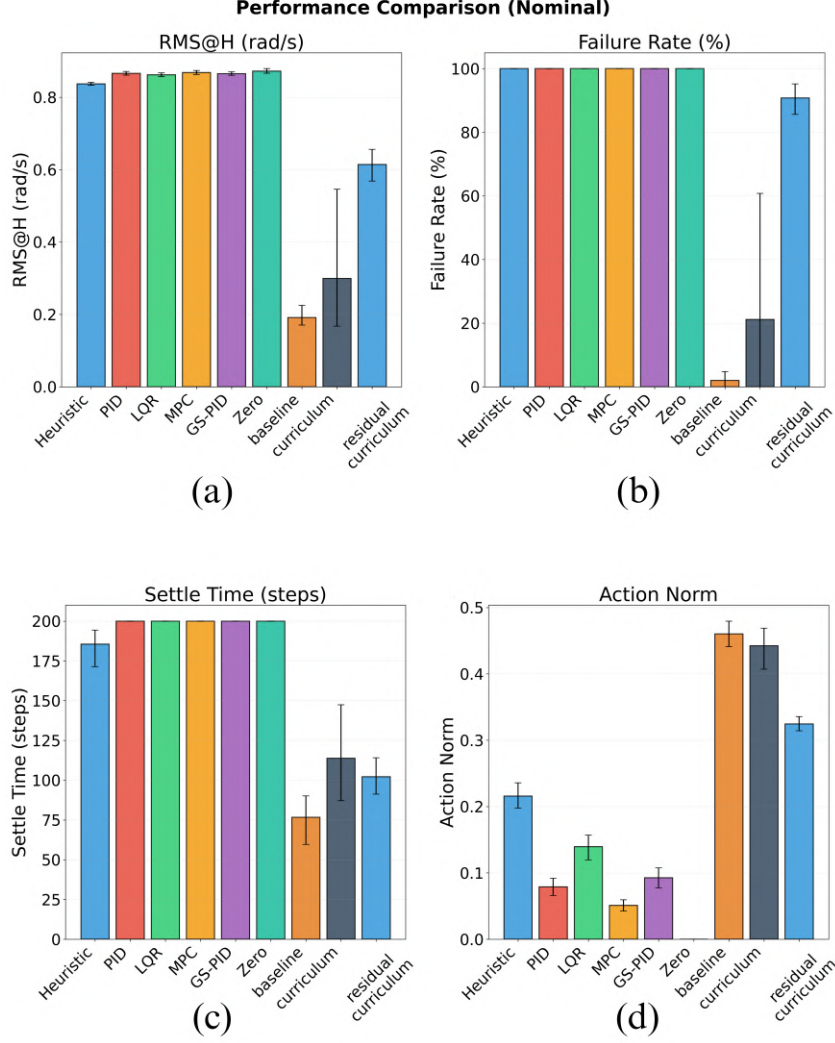


Fig. 2 Controller performance comparison under nominal physics. (a) Root Mean Square yaw error (RMS@H, in rad/s); (b) Failure Rate (%); (c) Settle Time (steps); (d) Action Norm, which measures average wing tip deflections magnitudes and serves as a proxy for agent control effort.

Residual curriculum (green) undergoes a catastrophic performance jump at approximately 500K steps, coinciding with the intermediate.twist phase transition, where RMS@H leaps from approximately 0.35 to 0.70 rad/s and never recovers. This is the training-side evidence for the 90.8% evaluation failure rate: the policy that was adequate at lower twist authority becomes destabilizing when the full dihedral channel is unlocked and the heuristic’s corrective outputs begin generating conflicting attitude torques.

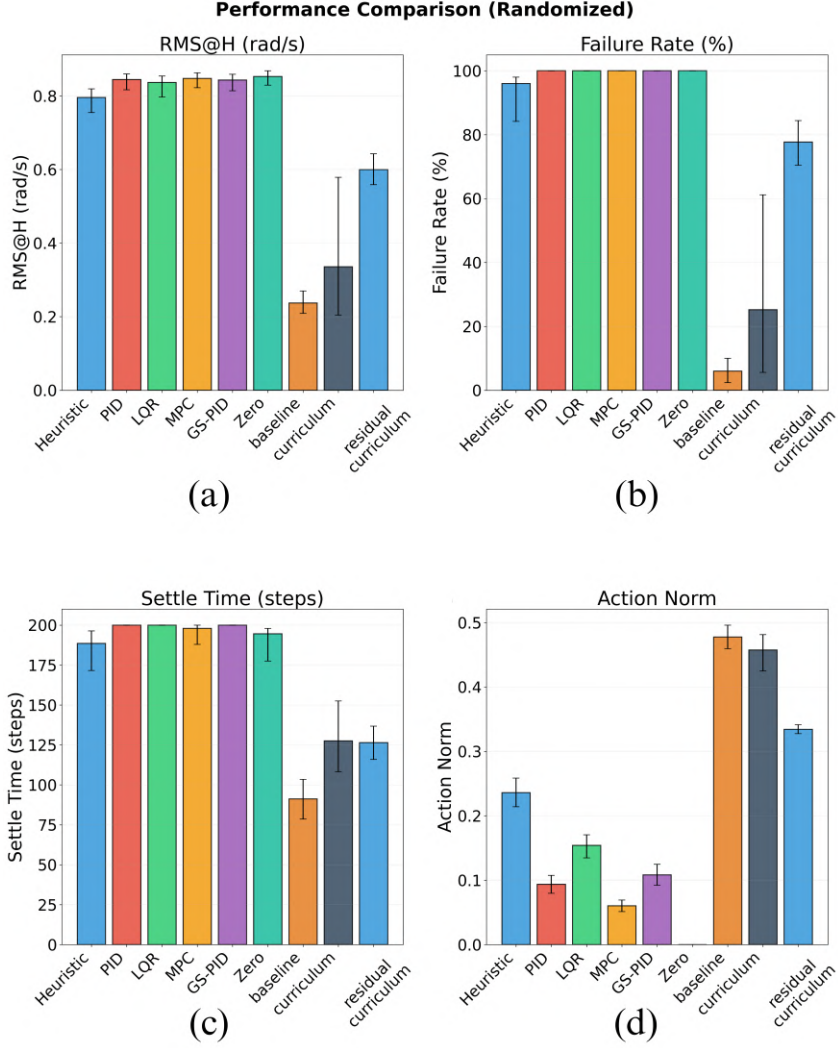


Fig. 3 Controller performance comparison under full domain randomization. (a) Root Mean Square Yaw error (RMS@H, in rad/s); (b) Failure Rate (%); (c) Settle Time (steps); (d) Action Norm.

4.4 Ablation Study

The ablation varies the evaluation-time domain randomization scale while holding the trained checkpoints fixed. Both conditions load the same five baseline SAC checkpoints, trained with `rand_scale` ramping to 1.0. As shown in Fig. 8, evaluating those checkpoints under nominal physics (`rand_scale` = 0.0) yields $\text{RMS@H} = 0.186$ with a 0.8% failure rate; evaluating under full randomization (`rand_scale` = 1.0) yields 0.227 and 7.9% failure. This is therefore a robustness test, not a training ablation, meaning that evaluation-time physics uncertainty degrades a policy trained with domain

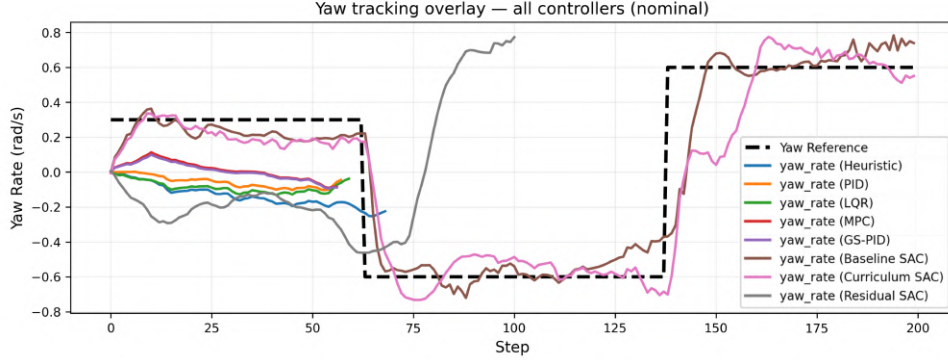


Fig. 4 Yaw rate step-response overlay across all controllers for a representative nominal episode.

randomization, not whether training with randomization was beneficial. The $\sim 22\%$ RMS degradation and $\sim 10\times$ failure rate increase under full randomization indicate a moderate robustness gap. A true training-ablation comparison — training a separate model with `rand_scale = 0` throughout — would be needed to quantify the benefit of domain randomization during training itself, and is left for future work.

4.5 Policy Interpretability Analysis

Policy interpretability is analyzed using five complementary lenses. (i) Fig. 9 reveals that `cos_pitch` (0.608) outranks `yaw_ref` (0.501) as the strongest driver of policy outputs, indicating an attitude-first control priority. (ii) The wing-asymmetry curve in Fig. 10 shows $R^2 = 0.001$ for the linear fit of Δz_{asym} against yaw reference — the policy uses asymmetry as a magnitude signal, not a direction signal. (iii) DAGger distillation to a KAN student in Fig. 11 achieves 0.259 RMS@H at iteration 15 with 33,240 parameters, a $2.3\times$ compression from the 78,000-parameter SAC actor; covariate shift causes loss increase from iteration 7 onward as the behaviour-cloning mixing coefficient β decays [13]. (iv) PCA of actor hidden activations in Fig. 12 shows PC1 separates the sign of the yaw target; only 2.1% of timesteps fall in the “tracking” regime while 40.5% are “stressed”, a separation further visualized by the latent t-SNE behavioral modes in Fig. 13. (v) As shown in Fig. 14, comparing the dihedral (Δz) action channels of baseline SAC and curriculum SAC reveals that both policies independently converge to the same morphing mechanism: differential dihedral modulation with a positive Δz bias on both wings, generating yaw moments through transient asymmetry.

5 Discussion

There are at least two plausible explanations for the curriculum’s lack of net benefit. First, the nine-term reward with survival bonus may already be well-shaped enough, providing sufficient gradient signal, for flat SAC to escape the $z = 0$ local optimum given sufficient timesteps — both methods receive the same total of approximately 6M steps, and the baseline simply uses them in one continuous phase rather than six

staged ones. Second, curriculum phase transitions themselves introduce instability: every time twist factor increments, the policy faces a shifted action space and reward landscape, which can destabilize policies that were near-converged. This is most visible in the residual variant but plausibly affects some curriculum seeds too.

The failure of Residual Heuristic Curriculum, where the agent achieved a 90.8% failure rate from attitude divergence, was caused by the additive heuristic and SAC outputs, which generated counterproductive non-linear attitude torques which neither controller originally intended. However, it is worth noting that this approach led to a steady-state RMS value of 0.140 among survivors where attitude did not diverge, the best of all controllers. This result confirms that the policy is sound when chaotic attitude is avoided.

As shown in Fig. 6 and Fig. 5, the failure mechanism is now well-understood from the attitude stability traces and the yaw overlay grid. During large yaw step transients, the SAC residual outputs large corrective commands that, when added on top of the heuristic’s simultaneous stabilization outputs, generate conflicting roll and pitch torques that neither controller originally intended. Roll grows steadily rather than being corrected, hitting the 65° termination limit at approximately $T = 101$ steps — visible in the yaw overlay grid as an abrupt early termination.

The aerodynamic proxy undershoots the inviscid VLM reference by approximately 31% (slope ratio = 0.687), which is intentional — viscous effects at $Re \approx 10^5$ suppress the real lift slope below the inviscid ceiling — but both models are unvalidated against physical hardware. Wind tunnel measurements at $Re \approx 10^5$ are required to confirm whether the proxy’s conservative calibration is accurate, or whether it represents a different error.

The structural model shares a similar gap: the Bézier curvature energy is a proxy for Euler-Bernoulli bending ($r = 0.9998$), but torsional stiffness, flex-torsion coupling, and aeroelastic flutter are not modelled. A three-point bend test on the physical spar would calibrate the actual bending stiffness EI, and a torsional stiffness measurement would determine whether the curvature-energy proxy adequately captures the coupled response.

At $n = 5$ seeds the achieved power is 0.20, and the minimum detectable effect at 80% power is $d \approx 1.75$. The curriculum null result ($d = -0.56$) cannot be confirmed or refuted without at least $n \approx 20$ seeds. All results in this paper are simulation-only; hardware validation on a physical glider is the critical next step, with domain randomization bounds requiring empirical calibration against flight test data.

6 Conclusion

This paper presented the successful implementation of a novel reinforcement learning framework, relying on a compliant cubic Bézier spar as the sole actuator, to achieve effective and precise yaw control in tailless flight. Notably, baseline SAC achieved a 2% failure rate and a $4.4\times$ accuracy improvement over the best-performing classical controllers, which suffered 96%–100% episode failure rates at the 65° roll/pitch evaluation limit due to the nonlinearity of the coupling of aerodynamic effects from

Bézier spar actuation. However, the null result of the curriculum SAC, whose performance was statistically equivalent to baseline SAC, as well as the negative result of the heuristic curriculum SAC, which generated conflicting roll and pitch torques that neither controller originally intended, resulting in high failure rates, are also worth reporting. The interpretability suite provided insight into each of the controllers, with the successful DAgger distillation to a KAN student revealing the baseline SAC and curriculum SAC’s successful attitude-first control priority, which suppressed failure rates to a minimum while maintaining stable yaw-control through differential dihedral modulation. Looking ahead, future work includes wind tunnel measurements at $Re \approx 10^5$ to properly calibrate the sim-to-real gap and translate simulation findings to hardware in real, physical flight.

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Per-controller yaw tracking

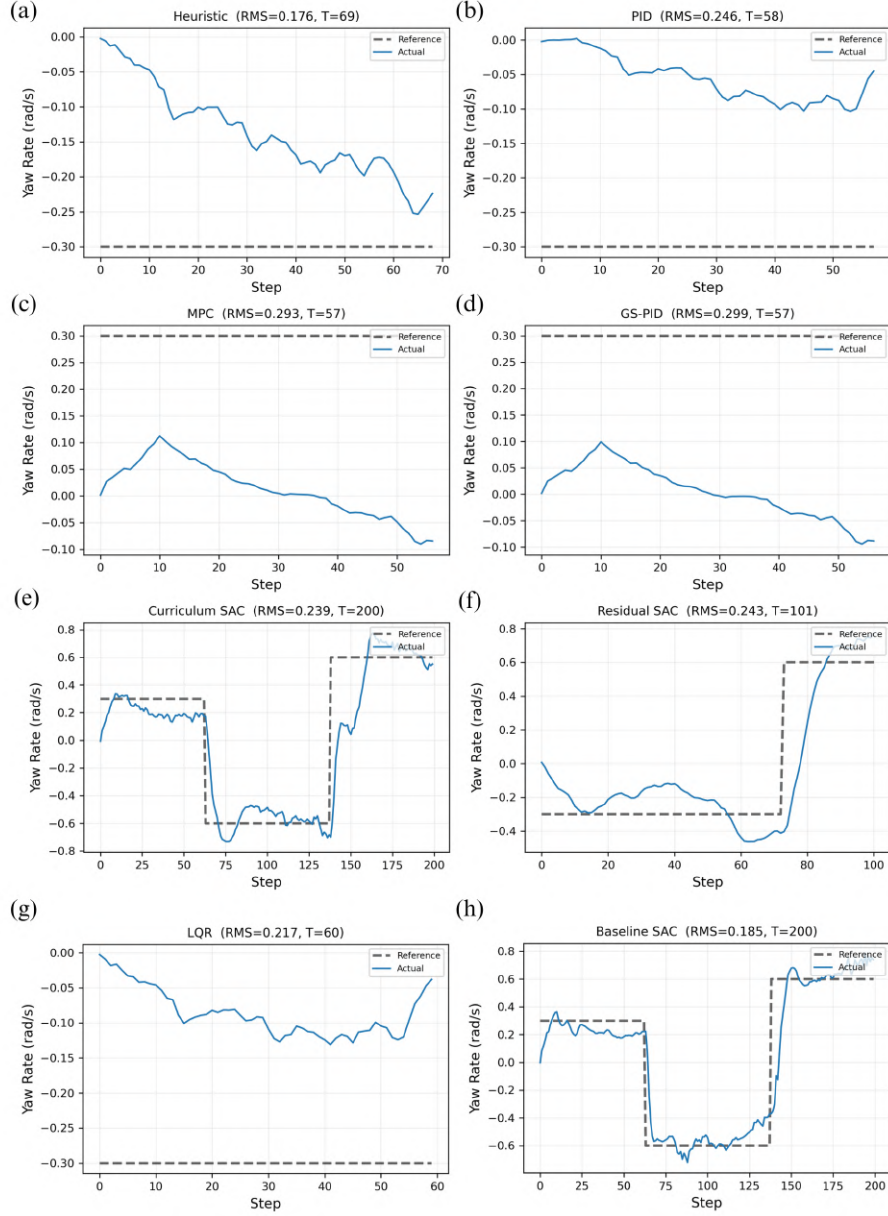


Fig. 5 Per-controller yaw tracking trajectories (best episode). Respective controllers: (a) Heuristic (RMS=0.176, T=69); (b) PID (RMS=0.246, T=58); (c) MPC (RMS = 0.293, T=57); (d) GS-PID (RMS=0.299, T=57); (e) Curriculum SAC (RMS = 0.239, T=200); (f) Residual SAC (RMS=0.243, T=101); (g) LQR (RMS=0.217, T=60); (h) Baseline SAC (RMS=0.185, T=200).

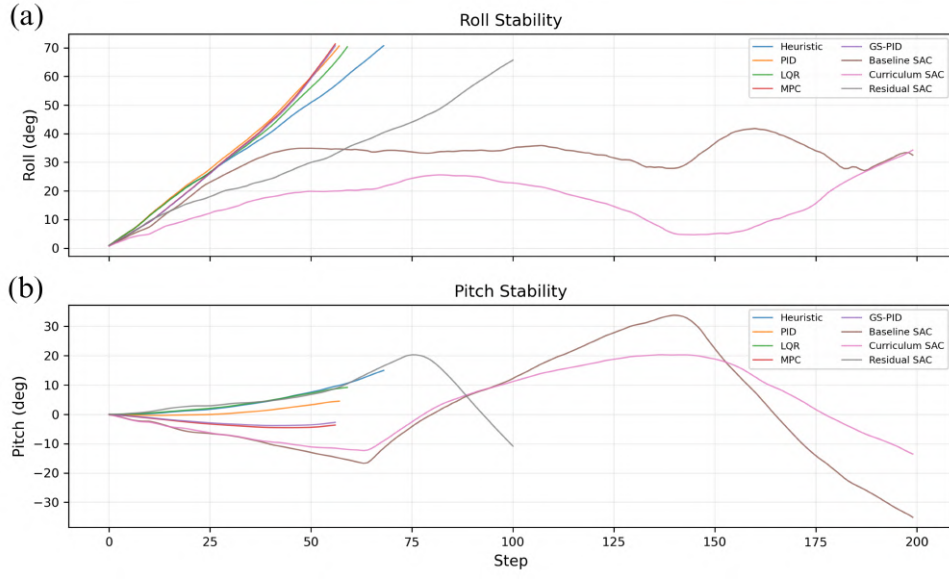


Fig. 6 (a) Roll and (b) pitch angle traces during evaluation episodes.

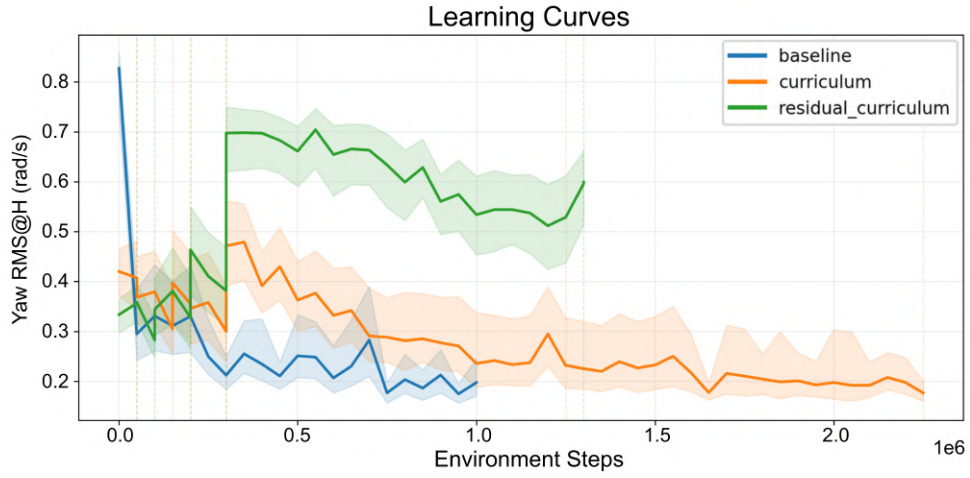


Fig. 7 Training learning curves (paper mode, 5 seeds). Residual curriculum (green) undergoes extreme performance drop coinciding with intermediate_twist phase. Baseline (blue) and curriculum (orange) demonstrate similar performance at the end of their training, with baseline converging faster.

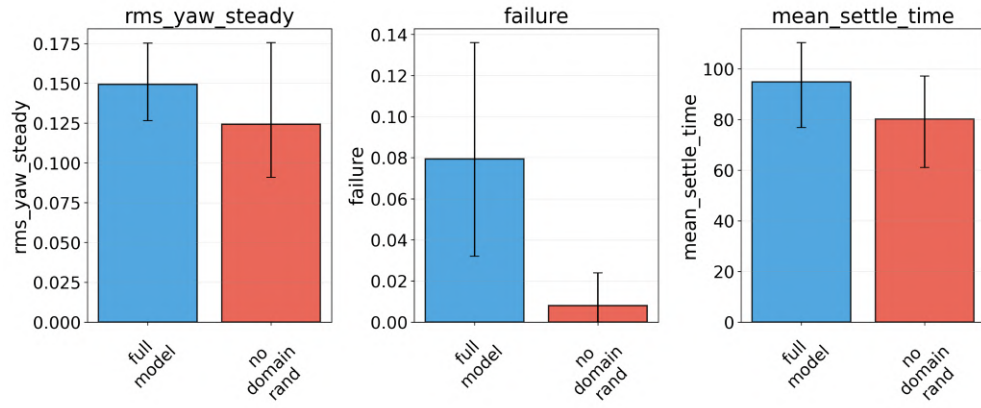


Fig. 8 Robustness of baseline SAC to evaluation-time physics uncertainty.

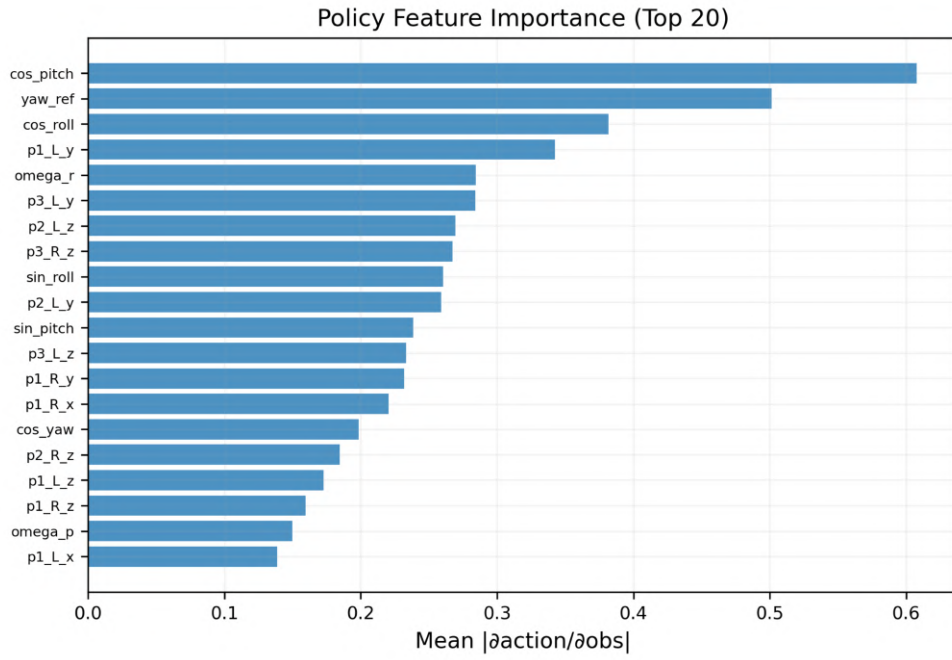


Fig. 9 Policy feature importance via finite-difference Jacobian (baseline SAC). Top feature is cos_pitch (0.608), not yaw_ref (0.501), indicating the policy prioritizes attitude maintenance over yaw tracking.

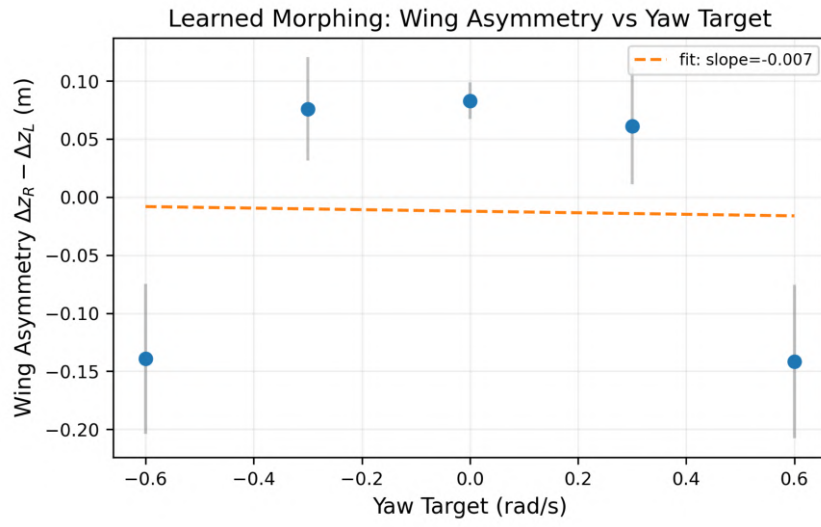


Fig. 10 Learned morphing strategy: wing z-asymmetry ($\Delta z_R - \Delta z_L$) vs. yaw target. Linear fit slope = -0.007, $r^2 = 0.001$. The policy does not use simple differential twist proportional to yaw demand; instead, a bimodal amplitude pattern is observed (extreme targets: negative asymmetry; intermediate: positive).

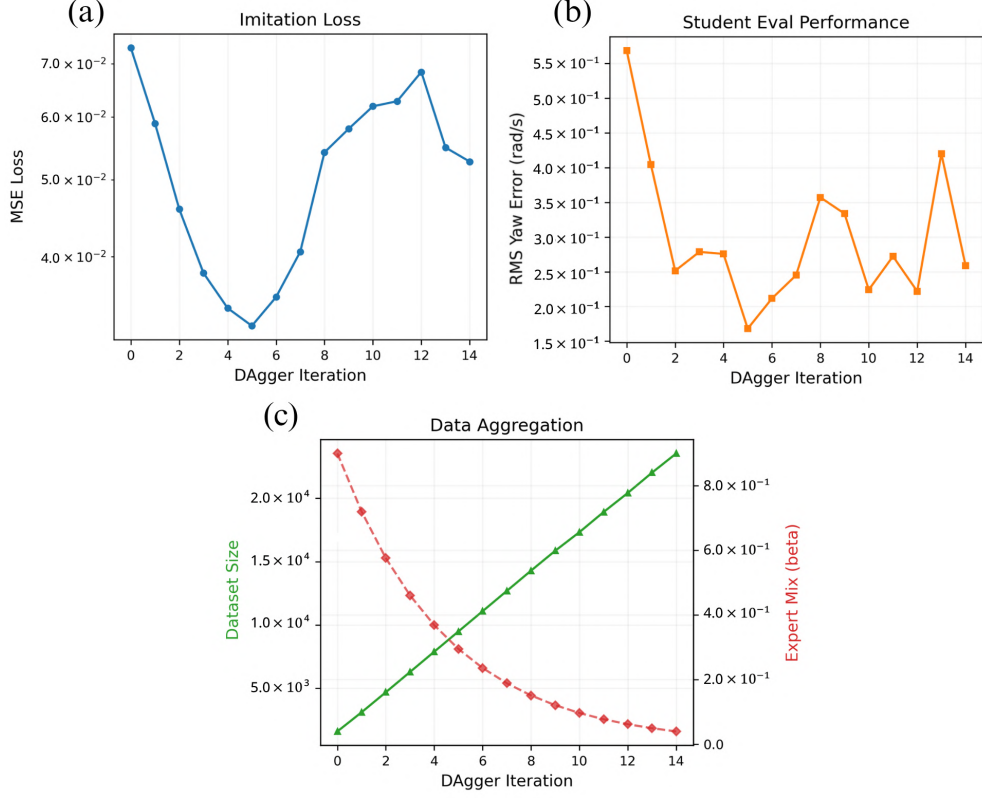


Fig. 11 DAGger imitation learning of KAN student from SAC expert. (a) Mean squared error loss vs. DAGger iterations, measuring how closely the SAC teacher’s actions matches the student policy’s output on the same observations. (b) Evaluation of student policy performance in terms of RMS Yaw Error (rad/s) over DAGger iterations, assessing student’s actual yaw-tracking abilities. (c) Dataset size (green, left y-axis) and expert mixing ratio β (red, right y-axis) over DAGger iterations. Dataset size is the total number of state-action transitions collected from both expert and student. β is the fraction of actions taken by the expert (SAC); it decays over iterations, exposing the student (KAN) to its own error distribution.

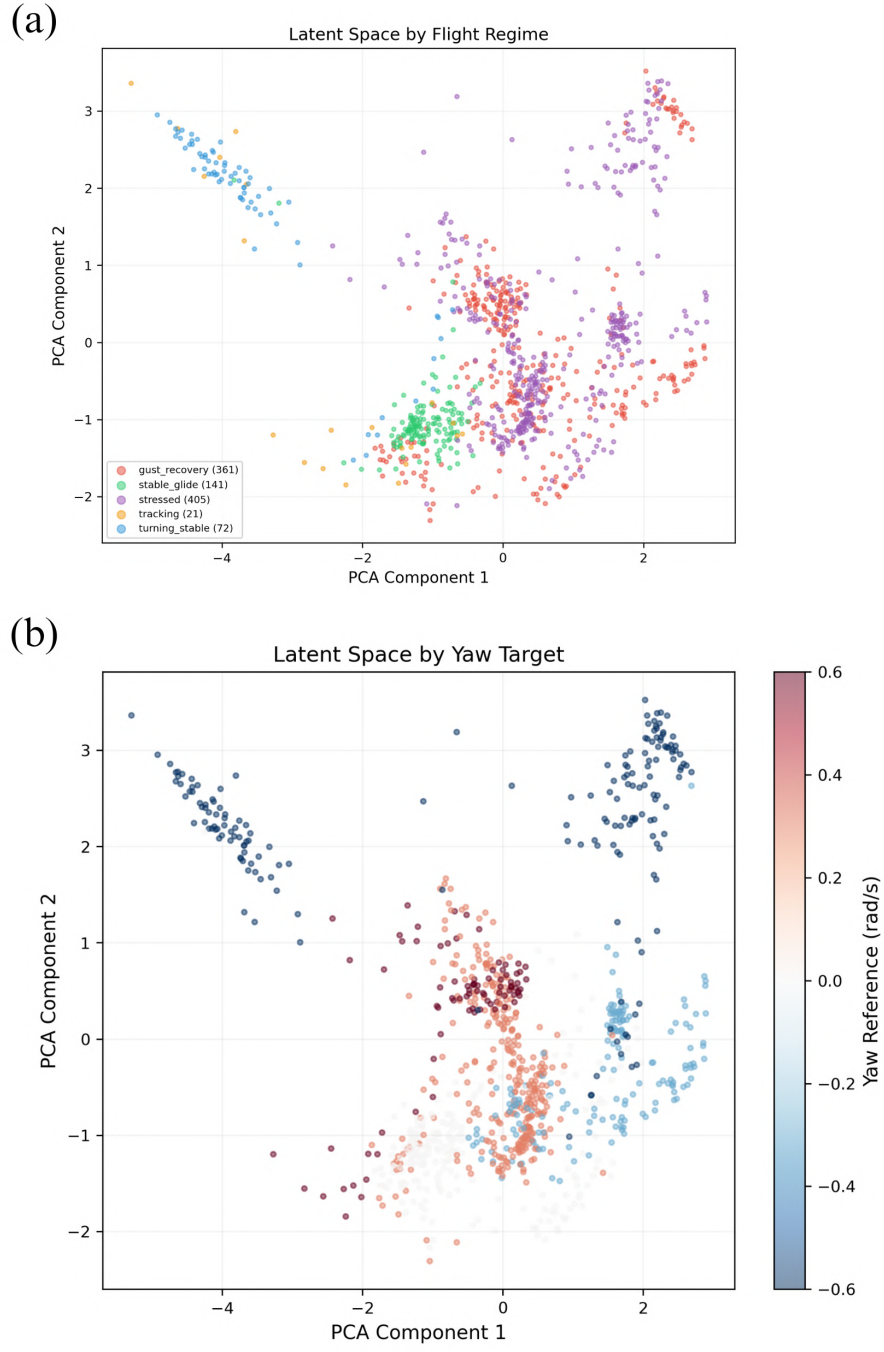


Fig. 12 Latent space PCA of baseline SAC actor activations ($n = 1000$ steps) separated by (a) flight regime and (b) yaw target.

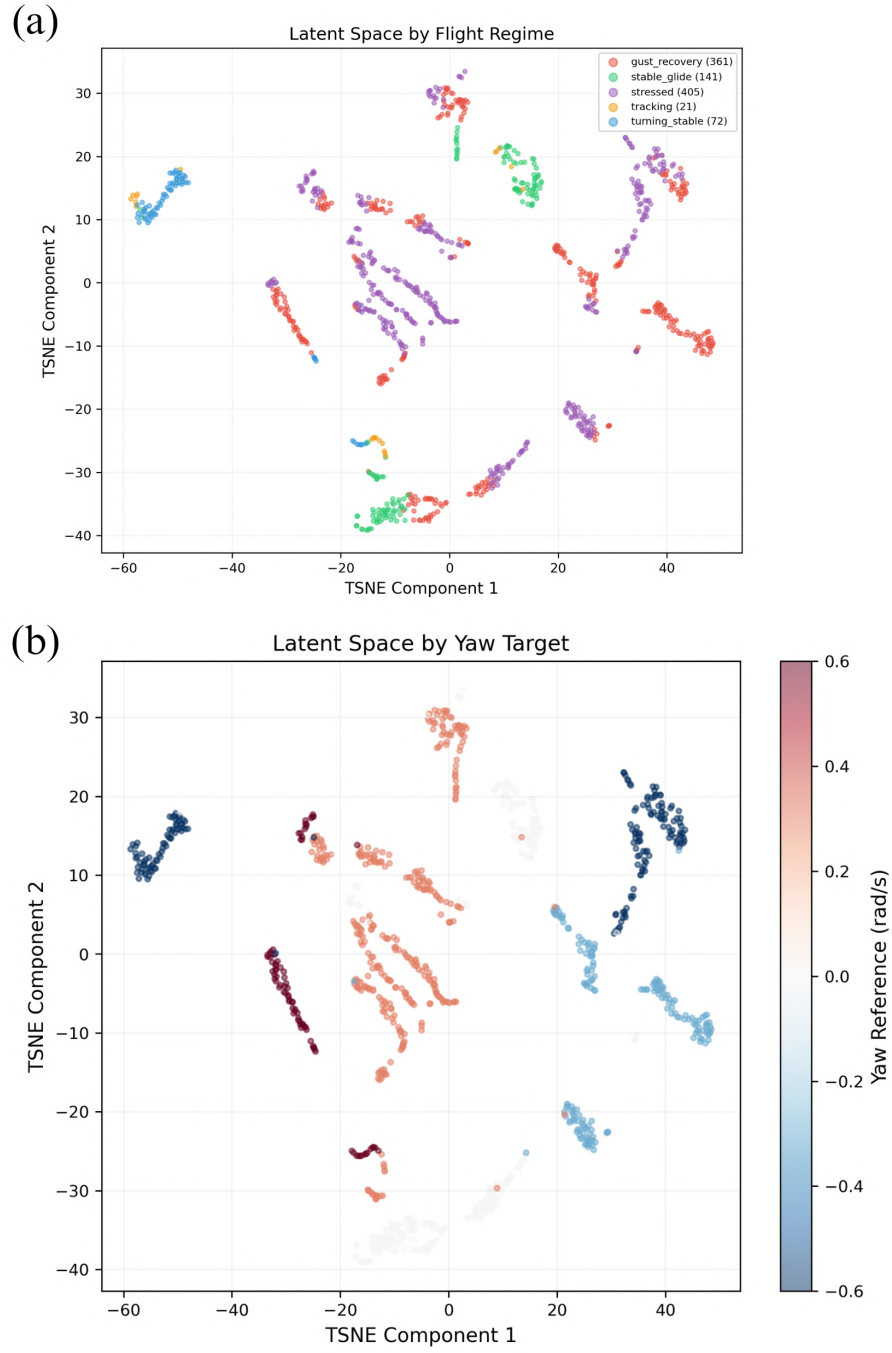


Fig. 13 Latent t-SNE of continuous evolution (trajectories) and distinct behavioral modes (islands), separated by (a) flight regime and (b) yaw target.

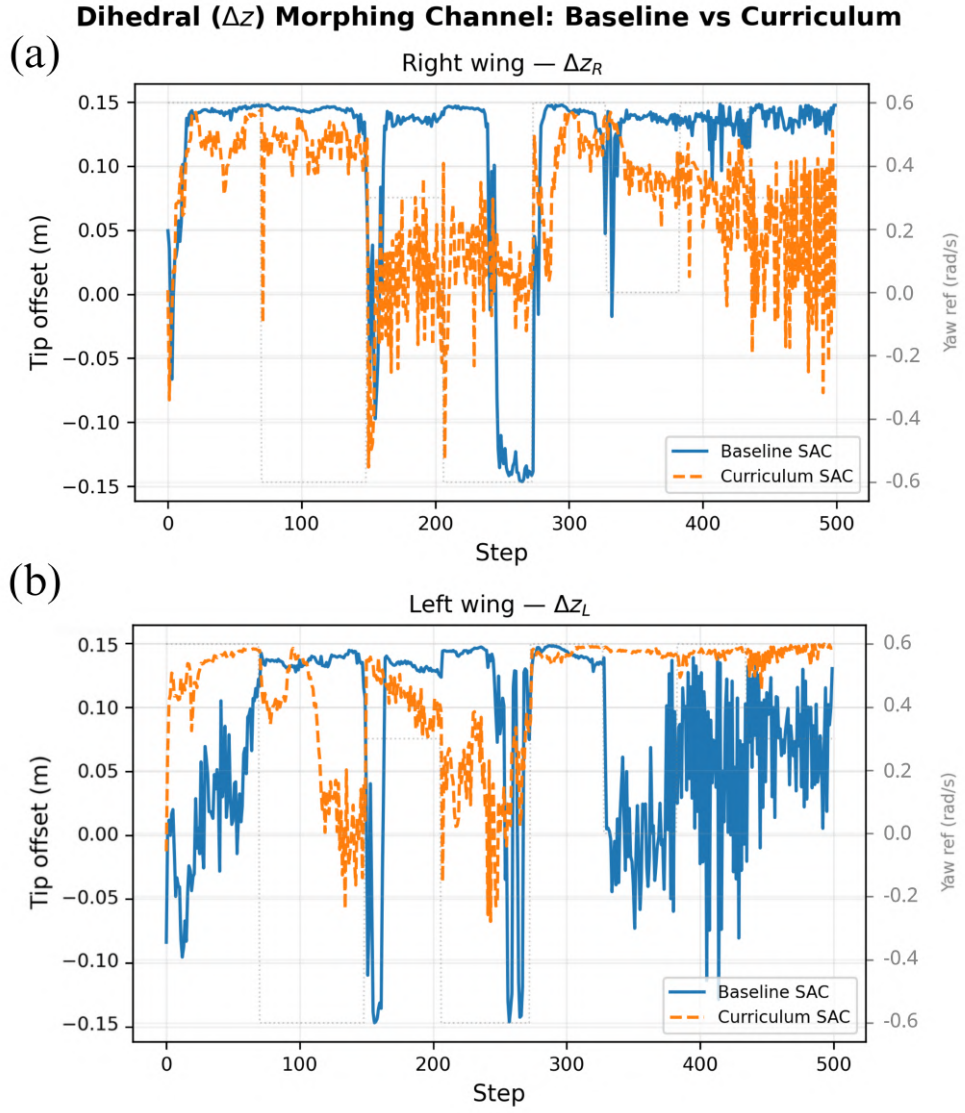


Fig. 14 Dihedral morphing channel of (a) right wing Δz_R and (b) left wing Δz_L for baseline SAC and curriculum SAC across a representative episode.